

SYNTHETIC TEST DOCUMENT — LIFE SCIENCES QUALITY ASSURANCE TEST DATASET — STRICTLY FOR DEMONSTRATION & VALIDATION

Hydrogenation Reactor Pressure Excursion Exceeding Operating Limits

1. GENERAL INCIDENT INFORMATION

Date of Occurrence:	2026-09-16	Discovery Source:	Process Historian / In-Line Transducer PT-302
Manufacturing Site:	API Synthesis Suite, Cork Manufacturing Plant	Department / Area:	Chemical Manufacturing
Related Product / Material:	Active Pharmaceutical Ingredient (API-709)	Batch / Lot Number:	LOT-88219

2. EVENT SUMMARY & DEVIATION DESCRIPTION

During catalytic hydrogenation of API-709 in Reactor R-201, nitrogen blanket pressure spiked to 4.8 bar for 18 minutes against the validated operating limit of 2.0 to 3.2 bar due to a malfunctioning exhaust back-pressure regulator valve (PRV-104).

3. IMMEDIATE ACTIONS & CONTAINMENT MEASURES

Emergency manual depressurization valve was opened by the shift technician to safely vent excess nitrogen to the scrubber line, restoring reactor pressure to 2.5 bar. Reactor R-201 was locked out and maintenance was notified.

4. INITIAL LOGGING VERIFICATION

Logged By: Operations Shift Lead Date: 2026-09-16	QA Reviewer: Quality Assurance On-Duty Date: 2026-09-16	Workflow Stage: Awaiting AI Extraction & Human Review
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